

Healthcare Multimodal Foundation Model System

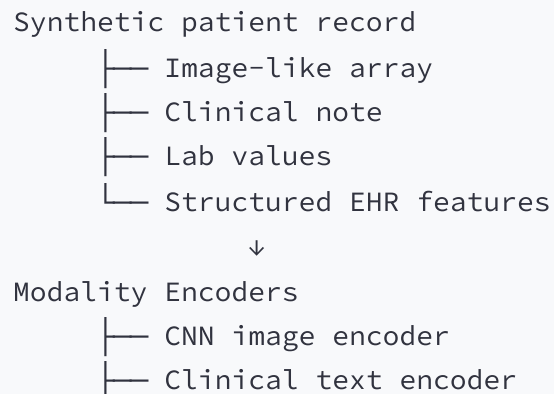
Image + clinical note + labs + structured EHR risk prediction

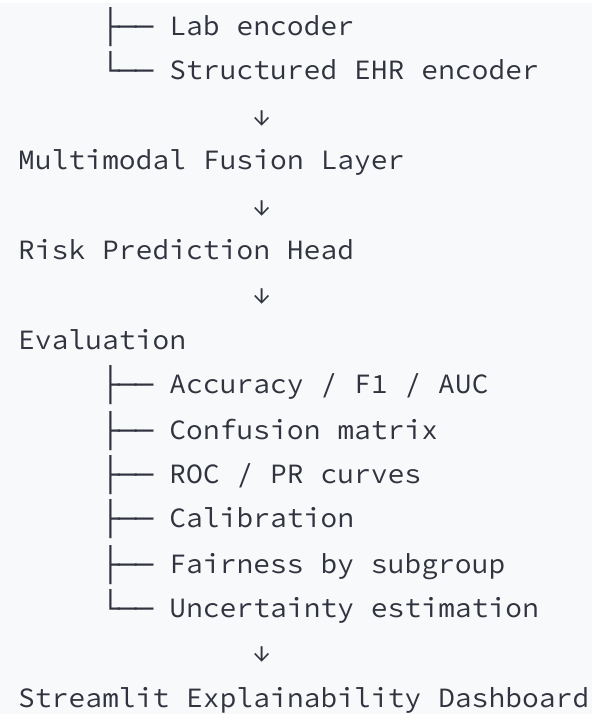
Project Question

Can multimodal healthcare AI combine medical images, clinical notes, labs, and structured EHR data to improve risk prediction and clinical interpretability?

Overview Metrics Fairness & Uncertainty Clinical Interpretation

Architecture





Production Upgrade Path

A production version could add:

- FHIR ingestion
- DICOM image ingestion
- ClinicalBERT or BioClinicalBERT
- Vision Transformer image encoder
- SHAP and Grad-CAM
- MLflow experiment tracking
- model registry
- external validation

- secure deployment